

SIMATS ENGINEERING

CO3-ASSESSMENT TOOL 1

Experiments

COURSE NAME: Computer Networks

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

Dr.V.Parthipan

Code of Conduct:

I B.Pardhu Arjun(192525245) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student : B.Pardhu Arjun

Criteria	Understandi ng of Concepts 2.5	Experimental Design 3	Use of Tools 2	Analysis of Results 1.5	Documenta tion 1	Total (10)
Q1	2.5	2.7	2	1.3	1	9.5
Q2	2.5	2.8	1.9	1.3	1	9.5
Q3	2.2	2.7	2	1.3	0.8	9
Q4	2.5	2.6	2	1.5	0.9	9.5

Faculty Signature

Experiments

QUESTIONS:4

Total Marks:40

1. Capture packets during a TCP connection setup. Identify the three packets involved in the handshake (SYN, SYN-ACK, and ACK). Demonstrate the role of each flag and how they establish a reliable connection. (BL4) (10)
2. Capture a DNS query using UDP in Wireshark. Compare the UDP header size with TCP and note the absence of sequence numbers or acknowledgments. Discuss how this lightweight design impacts speed and reliability. (BL4) (10)
3. Simulate packet loss (e.g., disconnect briefly) and capture traffic in Wireshark. Observe how TCP retransmits lost packets while UDP does not. Explain why this difference makes TCP reliable but slower compared to UDP. (BL4) (10)
4. Use Wireshark to capture DNS traffic while resolving a domain name. Identify the query type (A, AAAA, MX) and the response received. Measure the time taken for resolution and explain why DNS typically uses UDP. (BL4) (10)

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

SOLUTIONS:

1)

Wireshark - Packet 1276 - Wi-Fi

Frame 1276: Packet, 55 bytes on wire (440 bits), 55 bytes captured (440 bits) on interface \Device\NPF_{93CA38C7-E8C4-4430-9C4A-1457C602}

Ethernet II, Src: Intel_97:ec:21 (4c:0f:3e:97:ec:21), Dst: Cisco_6d:bc:f7 (24:d5:e4:6d:bc:f7)

Internet Protocol Version 4, Src: 172.23.23.169, Dst: 192.178.211.188

Transmission Control Protocol, Src Port: 60690, Dst Port: 5228, Seq: 1, Ack: 1, Len: 1

Data (1 byte)

Source Port: 60690
Destination Port: 5228
[Stream index: 4]
[Stream Packet Number: 1]
[Conversation completeness: Incomplete (60)]
[TCP Segment Len: 1]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 2686945166
[Next Sequence Number: 2 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)
Acknowledgment number (raw): 1067029538
0101 = Header Length: 20 bytes (5)

0000 24 d5 e4 6d bc f7 4c 0f 3e 97 ec 21 08 00 45 00 \$ - . L > . ! . E
0010 00 29 3f f6 40 00 80 06 00 00 ac 17 17 a9 c0 b2) 9 @
0020 d3 bc ed 12 14 6c a0 27 87 8e 3f 99 94 22 50 10 ... 1 " P
0030 00 fd 58 4b 00 00 00 .. XK ..

2)

Wireshark - Packet 1276 - Wi-Fi

Frame 1276: Packet, 55 bytes on wire (440 bits), 55 bytes captured (440 bits) on interface \Device\NPF_{93CA38C7-E8C4-4430-9C4A-1457C602}

Ethernet II, Src: Intel_97:ec:21 (4c:0f:3e:97:ec:21), Dst: Cisco_6d:bc:f7 (24:d5:e4:6d:bc:f7)

Internet Protocol Version 4, Src: 172.23.23.169, Dst: 192.178.211.188

Transmission Control Protocol, Src Port: 60690, Dst Port: 5228, Seq: 1, Ack: 1, Len: 1

Data (1 byte)

Source Port: 60690
Destination Port: 5228
[Stream index: 4]
[Stream Packet Number: 1]
[Conversation completeness: Incomplete (60)]
[TCP Segment Len: 1]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 2686945166
[Next Sequence Number: 2 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)
Acknowledgment number (raw): 1067029538
0101 = Header Length: 20 bytes (5)

0000 24 d5 e4 6d bc f7 4c 0f 3e 97 ec 21 08 00 45 00 \$ - . L > . ! . E
0010 00 29 3f f6 40 00 80 06 00 00 ac 17 17 a9 c0 b2) 9 @
0020 d3 bc ed 12 14 6c a0 27 87 8e 3f 99 94 22 50 10 ... 1 " P
0030 00 fd 58 4b 00 00 00 .. XK ..

3)

The image shows a Wireshark packet capture of a TCP connection. The packet list on the left shows several packets, with packet 677 highlighted. The packet details pane on the right shows the structure of packet 677, which is a TCP Spurious Retransmission. The packet is a TCP segment with sequence number 443, destination port 64182, and acknowledgment number 1732. The packet is 728 bytes long. The packet is a retransmission of a previous segment, as indicated by the 'Spurious Retransmission' label. The packet is captured on the interface 'Wi-Fi'.

Packet 677: TCP Spurious Retransmission [443 → 64182 [PSH, ACK] Seq=4237 Ack=1732 Win=268032 Len=670 [TCP PDU reassembled in 673]]

Transmission Control Protocol, Src Port: 443, Dst Port: 64182, Seq: 4237, Ack: 1732, Len: 670

Source Port: 443
Destination Port: 64182
[Stream index: 1]
[Stream Packet Number: 13]
[Conversation completeness: Incomplete, DATA (15)]
[TCP Segment Len: 670]
Sequence Number: 4237 (relative sequence number)
Sequence Number (raw): 3647738919
[Next Sequence Number: 4907 (relative sequence number)]
Acknowledgment Number: 1732 (relative ack number)
Acknowledgment number (raw): 2506217217
0101 = Header Length: 20 bytes (5)
Flags: 0x018 (PSH, ACK)
Window: 1047
[Calculated window size: 268032]

Frame 677: Packet, 728 bytes on wire (5824 bits)
Ethernet II, Src: Cisco6d:bc:f7 (24:d5:e4:6d:bc:f7), Dst: Intel97:ec:21 (4c:0f:3e:97:ec:21)
Internet Protocol Version 4, Src: 142.251.221.206, Dst: 172.23.23.169
Transmission Control Protocol, Src Port: 443, Dst Port: 64182, Seq: 4237, Ack: 1732, Len: 670
Source Port: 443
Destination Port: 64182
[Stream index: 1]
[Stream Packet Number: 13]
[Conversation completeness: Incomplete, DATA (15)]
[TCP Segment Len: 670]
Sequence Number: 4237 (relative sequence number)
Sequence Number (raw): 3647738919
[Next Sequence Number: 4907 (relative sequence number)]
Acknowledgment Number: 1732 (relative ack number)
Acknowledgment number (raw): 2506217217
0101 = Header Length: 20 bytes (5)
Flags: 0x018 (PSH, ACK)
Window: 1047
[Calculated window size: 268032]

4)

The image shows a Wireshark packet capture of a DNS query and response. The packet list on the left shows several packets, with packet 1970 highlighted. The packet details pane on the right shows the structure of packet 1970, which is a DNS query response. The packet is a DNS response with transaction ID 0x8ce7, source port 53, and destination port 58963. The packet is 212 bytes long. The packet is captured on the interface 'Wi-Fi'.

Packet 1970: DNS Standard query response 0x8ce7 A mobile.events.data.microsoft.com CNAME mobile.events.data.trafficmanager.net CNAME onedscolorprdeus19...

Domain Name System (response)

Transaction ID: 0x8ce7
Flags: 0x1800 Standard query response, No error
Questions: 1
Answer RRs: 3
Authority RRs: 0
Additional RRs: 0

Queries
mobile.events.data.microsoft.com: type A, class IN

Answers
[Request ID: 1969]
[Time: 20.529700 milliseconds]

Frame 1970: Packet, 212 bytes on wire (1696 bits), 212 bytes captured (1696 bits) on interface \Device\NPF_{93CA30C7-E0C4-4430-9C4A-1457C60276E2}, id 0
Ethernet II, Src: Cisco6d:bc:f7 (24:d5:e4:6d:bc:f7), Dst: Intel97:ec:21 (4c:0f:3e:97:ec:21)
Internet Protocol Version 4, Src: 172.21.100.31, Dst: 172.23.23.169
User Datagram Protocol, Src Port: 53, Dst Port: 58963
Domain Name System (response)